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Visualization and numerical simulation of supersonic microjets

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Abstract. Very narrow supersonic jets expanding from a small size convergent-divergent nozzle are visualized by the laser induced fluorescence method and simulated numerically using the piecewise linear method. Good agreement between the experiment and the numerical result is obtained in the jet structure, i.e. the shape of the barrel shock, the location of its reflection point on the axis, and the shape of the jet boundary.

1 Introduction

Small-size Laval nozzles have some advantages in an efficient production of clusters condensed from the expanding gas, in the concentration of high speed flows within a very small angle, which will be useful in direction control of propulsion systems, in the acceleration and the transportation of small particles in space and also in the production of high intensity molecular beams.

In the present paper we report visualization and numerical simulation of a microjet issued from such a small-size Laval nozzle. For the visualization we use the laser induced fluorescence (LIF) method (Smalley et al. 1976; Hiller & Schmidt-Ott 1977; Teshima et al. 1984; Teshima & Nakatsuji 1984), which can be used for a wide density range from high density, where the optical interferometry is still effective, to low density, where electron beam excitation or electric discharge is usually applied. Numerical simulations are made using the piecewise linear method (PLM). It has been proved that this method has very small smearing effects with practically no oscillations in its solution near discontinuities such as shocks and contact surfaces (Woodward & Colella 1981; Saito et al. 1986).

2 Flow visualization by laser induced fluorescence

Iodine molecules can be excited by an argon ion laser of 514.5 nm to states which fluoresce at longer wavelengths. The strength of LIF is strong enough to allow us to visualize the flow to naked eyes with a small amount of iodine molecules seeded in the flow medium at its vapor pressure (about 30 Pa at room temperature), which does not disturb the flow itself, and with a laser power of 0.3–1.0 W.

The strength of LIF relates to the power of the laser, the number density in the lower state of the absorption line, its absorption coefficient and the transition probability of the excited state. The absorption coefficient and the number density of the lower state depend on the pressure, density and temperature of the flow. Therefore, the shock waves and the jet boundary, where the flow properties change largely, can be seen as differences in the strength of LIF. Since we use a longitudinally multi-mode laser, which includes many distinct resonance lines, the strength of LIF also depends on the laser lines mainly being absorbed. The output power is maximum at the frequency center of the laser, ν_{0_l}, and it decreases with increasing frequency difference from it. The absorption lines are mainly P13 and R15 lines in the (43-0) band of B-X system of iodine, which have the same unresolved frequency lower than ν_{0_l} (Gerstenkorn & Luc 1979). Therefore, the iodine molecules with a velocity component, which increases the frequency of the absorption lines by the Doppler effect, absorb stronger laser lines, and vice versa. Consequently, the strength of LIF becomes unsymmetrical with respect to the jet axis.

A schematic view of the experimental apparatus is shown in Fig. 1. Supersonic jets are obtained by expanding the gas through a Laval nozzle after flowing through an iodine cell. Since iodine molecules have very aggressive properties, most of the I_2-vapor is collected by a cold trap mounted between the chamber and the vacuum pump. The jet was irradiated by the laser beam focused on the jet axis at right angles with the axis, and the resultant LIF was recorded photographically by sweeping the laser beam along the jet axis at the direction perpendicular to both the jet axis and the laser beam. The laser (NEC GLG 3200) was operated at a laterally single- and longitudinal multi-mode. The Laval nozzle had a 30° half-

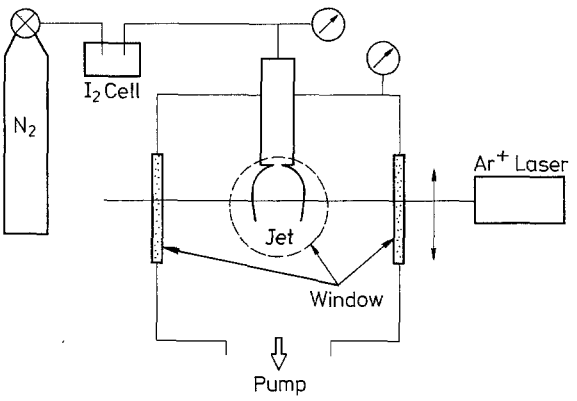

Fig. 1. Schematic view of experimental apparatus

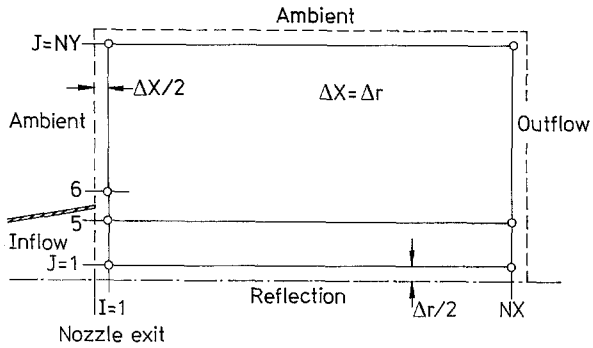

Fig. 2. Computational domain and boundary conditions

angle and 3.5 mm-long conical convergent section and a 10° full-angle and 8.5 mm-long conical divergent section with a 0.5 mm throat diameter. The area ratio of the nozzle exit to the throat section was 15.7. Nitrogen gas at room temperature was used at the stagnation pressure, p_0, of 1–4 bar. In order to obtain the pressure ratios, $p_0/p_\infty = 500$, 1,000, 1,500 and 2,000, the pressure in the vacuum chamber p_∞ was changed accordingly.

3 Numerical simulation by the piecewise linear method

The governing equation for an axisymmetric flow of an inviscid-compressible gas without heat conduction can be written as follows:

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial r} + Z = 0, \tag{1}$$

where

$$U = \begin{pmatrix} \varrho \\ \varrho u \\ \varrho v \\ \varrho E \end{pmatrix}, \quad F = \begin{pmatrix} \varrho u \\ \varrho u^2 + p \\ \varrho u v \\ u(\varrho E + p) \end{pmatrix}, \quad G = \begin{pmatrix} \varrho v \\ \varrho u v \\ \varrho v^2 + p \\ v(\varrho E + p) \end{pmatrix},$$

$$Z = \frac{1}{r} \begin{pmatrix} \varrho v \\ \varrho u v \\ \varrho v^2 \\ v(\varrho E + p) \end{pmatrix},$$

and t is time, x and r are axial and radial distances, p and ϱ are pressure and density of the flow, u and v are the velocities in x and r directions, respectively, and E is the energy of the gas per unit mass which can be written as

$$E = \frac{1}{\gamma - 1} \frac{p}{\varrho} + \frac{1}{2}(u^2 + v^2)$$

for a perfect gas with specific heat ratio γ.

Table 1. Conditions at nozzle exit. a is speed of sound and suffix 0 denotes stagnation conditions

Mach number			4.40		
pressure			$0.39 \times 10^{-2} p_0$		
density			$0.19 \times 10^{-1} \varrho_0$		
velocity in r direction v_J/a_0					
$J=$	1	2	3	4	5
	0.0087	0.026	0.044	0.061	0.078

Applying the operator splitting method to Eq. (1), the solution, W^{n+1}, at the time step $t + \Delta t$, can be obtained by the three one-dimensional sweeps in terms of the solution, W^n, at time t as

$$W^{n+1} = [L_x(\Delta t/2)\, L_r(\Delta t)\, L_x(\Delta t/2)]\, W^n.$$

Each set of the one-dimensional equations was solved numerically using the PLM. The details of the scheme is described by Woodward and Colella (1981) and Saito et al. (1986). The computational domain and the boundary conditions are shown in Fig. 2. We took 30 meshes in r-direction including 5 meshes as the radius of the nozzle exit and in x-direction 150, 180 and 210 meshes were taken for the pressure ratios of 500 and 1,000, 1,500, and 2,000, respectively. At the nozzle exit the flow properties, calculated from the stagnation condition and the isentropic relations using the geometrical area ratio, were given. We assumed that the boundary layer effect was negligible, because the observed shape of the jet did not change significantly by changing the stagnation pressure. We also assumed that the radial velocity at the nozzle exit was given so as the flow expanded radially along the divergent angle of the nozzle. These values on the grid points at the nozzle exit used for the calculation are given in Table 1. Outside of the nozzle exit at the left edge and on the outer edge of the computational domain the ambient conditions (p_∞, ϱ_∞, T_0 and $u = v = 0$) were given as boundary conditions. At the downstream right end outflow conditions and on the jet axis reflection conditions were given as boundary conditions. At $t = 0$ we set the

same conditions as those of the nozzle exit to the mesh points inside the region where r is smaller than the radius of the nozzle exit, and gave the ambient conditions to the rest of the mesh points. The calculation was made until a stationary flow was obtained.

4 Numerical results

The flow properties for $\gamma = 1.4$ and $p_0/p_\infty = 1{,}000$ are shown in Fig. 3 as a typical numerical result. For this case 750 steps were necessary to obtain a converged solution, which required 25 minutes with FACOM M382 at the Data Processing Center of Kyoto University. The distances in the figure are normalized with the radius of the nozzle exit, r_e, The density and the pressure contour lines are shown in Fig. 3a and b, where their total range is divided equally into 30 in a logarithmic scale. Close to the axis the density and pressure decrease rapidly until the barrel shock, which starts from the edge of the nozzle exit, reflects at the axis. Then the barrel shock expands again and interacts with the jet boundary where the diameter of the jet becomes a minimum. Different from the free jet expanding from a sonic orifice, the normal shock (Mach disk) does not exist. As can be seen in Fig. 3c showing the velocity vectors, the flow continues to expand after the nozzle exit but is suppressed by the converging barrel shock and then expands again after the point of shock reflection. The velocity decreases slightly at this point but still remains supersonic thereafter. The Mach number on the axis increases from the exit value, 4.40, to a maximum value just before the shock reflects. It takes a minimum value at the point where the diameter of the jet becomes minimum and increases again with the further expansion. Since the velocity on the axis is almost constant, this change of the Mach number is mainly attributed to the temperature change.

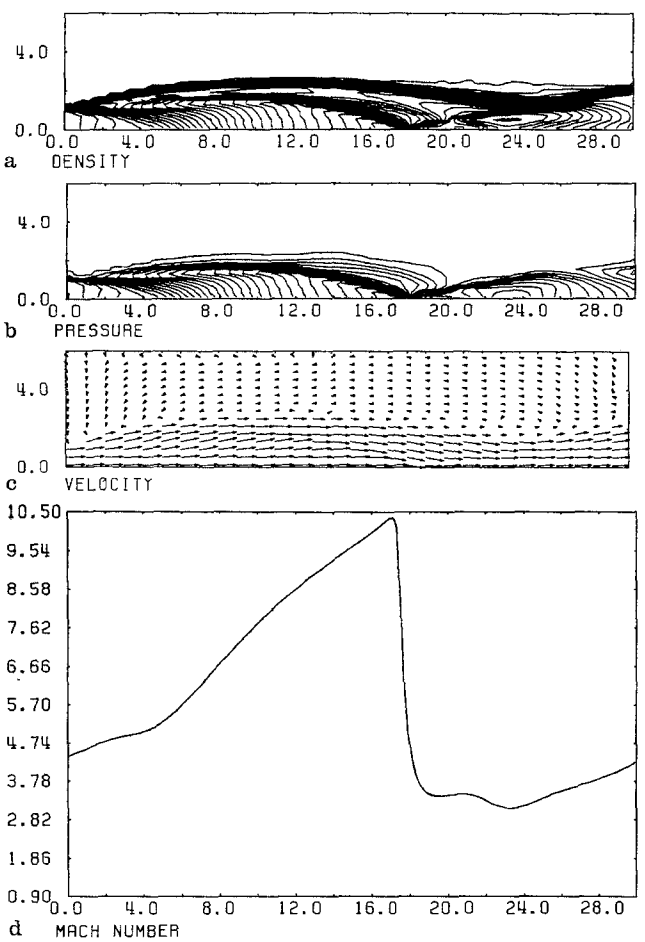

Fig. 3a–d. Numerical results for $\gamma = 1.4$ and $p_0/p_\infty = 1{,}000$. Distances x and r are normalized with radius of nozzle exit. **a** Density contour lines, **b** pressure contour lines, **c** velocity vectors, **d** Mach number change on axis

5 Comparison between visualization and numerical simulation

As is described in Chap. 2, the strength of LIF depends on the thermodynamic states of the flow. If we assume that the absorption coefficient is independent of temperature and pressure, then the strength of LIF is proportional to the number density in the lower states of the absorption lines of iodine molecule. Assuming the rotational temperature of the iodine molecules being equal to the translational temperature of the flow, the relative strength of LIF can be expressed as a function of the density and temperature of the flow. Then we can calculate the relative strength of LIF in the whole flow field using the numerical results. In this calculation, however, quenching effects, which also depend on the density and temperature, and Doppler effects are not included. Comparing the numerically obtained distribution of the relative strength of LIF with the photographs shows that the numerical computation is appropriate to simulate the jet structure quite well (Fig. 4). The unsymmetry in the experimentally obtained LIF intensity with respect to the jet axis, is a result of the Doppler effect, which was not included in the calculation.

When the temperature of iodine molecules decreases so that the Doppler width of the absorption lines becomes sufficiently narrow, the absorption of the laser by the iodine molecules only occurs by the Doppler tuning, i.e. the laser is absorbed discontinuously along the beam, and the resultant LIF represents stripes in the flow field (Teshima & Nakatsuji 1984). These stripes can be seen in the photographs for the cases of the pressure ratios 1,500 and 2,000. Since the velocity is almost constant in the region surrounded by the shock, these stripes represent the directions of flow. Comparisons of the photographs with the calculated velocity vectors are shown in Fig. 5. It can be seen that the directions of the flow and the stripes also agree well.

As the representative values for the jet size we may take the maximum diameter, x_t, of the jet and the dis-

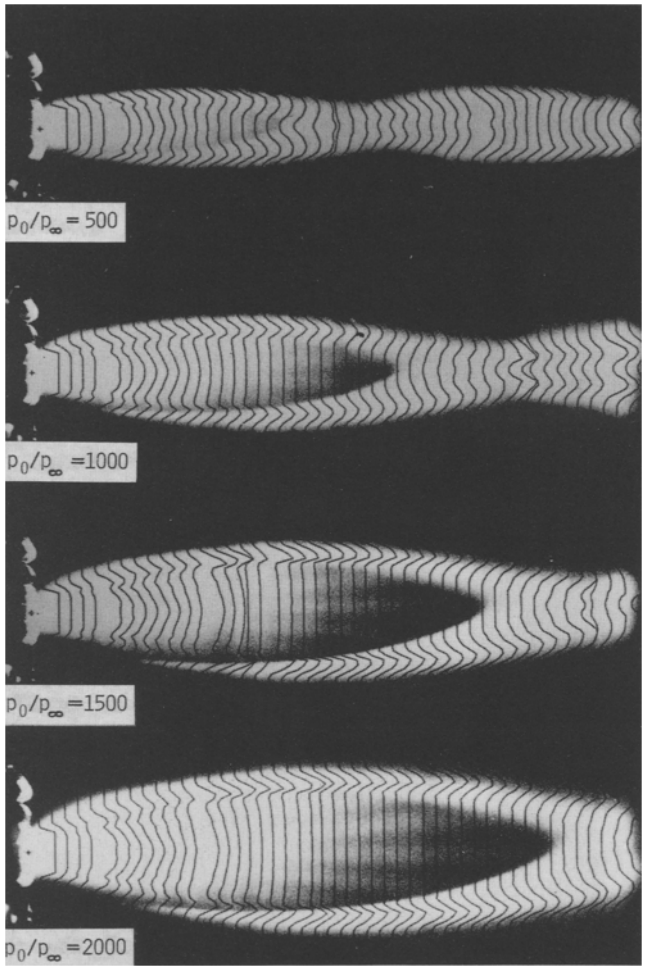

Fig. 4. Comparison between visualized jets and numerical simulation at various values of p_0/p_∞

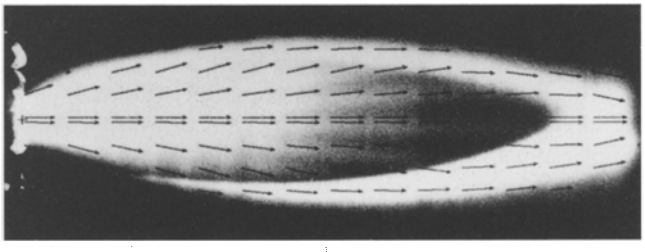

Fig. 5. Comparison between visualized flow directions and calculated velocity vectors for $p_0/p_\infty = 2{,}000$

tance, x_s, of the shock reflection point measured from the nozzle exit. These values obtained from the experiment and the calculation are compared in Fig. 6, showing also a very good agreement.

6 Conclusions

Supersonic jets issued from a small Laval nozzle with large pressure ratios were visualized and numerically simulated. The structure of the jet can be clearly seen by the LIF method. The numerical result is useful to understand the details of the flow. Good agreement between the experiment and the numerical results was obtained in the jet structure, i.e. the shape of the barrel shock and the location of its reflection point on the axis and also the shape of the jet boundary.

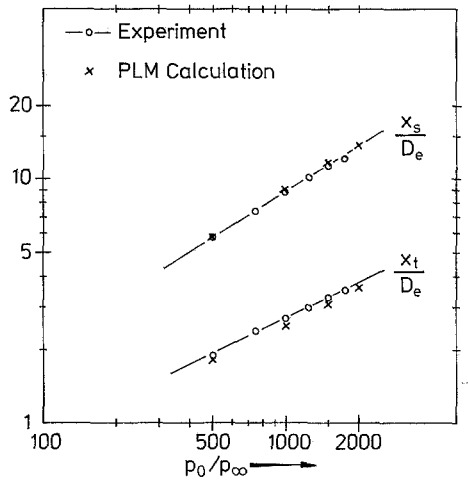

Fig. 6. Maximum jet diameter, x_t, and location of shock reflection point, x_s, normalized with diameter of nozzle exit, D_e, obtained experimentally and numerically as a function of p_0/p_∞

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